

Too Far Behind? Narrowing the Gap with a Dual-Enhanced Two-Stage Algebraic Framework for LWE

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Abstract. The Learning with Errors (LWE) problem forms the foundation for numerous post-quantum cryptographic schemes, such as the NIST-selected CRYSTALS-KYBER and CRYSTALS-DILITHIUM. Algebraic analysis on LWE traditionally relies on solving the Arora-Ge system via Gröbner bases, yet its performance is far from satisfactory when only a limited number of samples is available. Meanwhile, recent dual attacks have proven highly effective against concrete LWE-based algorithms. This gap motivates us to investigate whether integrating the technique from dual attacks into algebraic analysis can have a positive effect.

We propose a novel, two-stage algebraic algorithm for LWE. First, dual lattice reduction is applied to transform the original samples into a reduced dimension. From an algebraic perspective, this stage reduces the number of variables at the cost of increasing the noise. Second, instead of solving the classic Arora-Ge system, we introduce a new polynomial construction that exploits the error distribution and solves it via a resultant-based method. When given $m = n$ samples, our two-stage algorithm yields better complexity estimates for CRYSTALS-KYBER than Gröbner bases reported in Steiner’s work (Eurocrypt 2024). As an independent contribution, we show that for LWE with a small secret, applying the resultant-based method directly to the Arora-Ge system provides a provable complexity estimate that achieves an exponential speed-up over the proven bounds established by Steiner.

Finally, we show how various forms of side information—namely, perfect hints, modular hints, and approximate hints—can be systematically incorporated into our two-stage algorithm.

Keywords: Learning with Errors · Algebraic analysis · Dual lattice reduction · Gröbner bases · Resultant · CRYSTALS-KYBER · Hints

1 Introduction

The proposal of the Shor algorithm in 1994 shows that cryptographic algorithms based on integer factorization can be solved in polynomial time on a quantum

computer. This has led to a significant shift toward developing post-quantum cryptography (PQC) over the past few decades. PQC refers to cryptographic algorithms designed to remain secure in the case of both classical and quantum adversaries. Among these, lattice-based algorithms have attracted considerable attention. To date, the National Institute of Standards and Technology (NIST) has announced five standardization schemes for PQC, three of which are lattice-based. Among these, CRYSTALS-KYBER [31] and CRYSTALS-DILITHIUM [21] are directly constructed on the Learning with Errors (LWE) problem, which was introduced by Regev [28]. Therefore, a thorough understanding of the concrete complexity of LWE is crucial, motivating the cryptanalysis of LWE from diverse perspectives.

In this paper, two analytical approaches for LWE are considered. The first is **algebraic analysis**, proposed by Arora and Ge in ICALP 2011 [4]. By exploiting the fact that each error coordinate is bounded in an integer interval $\llbracket -T, T \rrbracket$, they construct a polynomial system regarding the secret vector, where each polynomial has a degree $d = 2T + 1$. The system is subsequently solved using a linearization technique. In ACISP 2020, Sun et al. [33] studied the linearization technique specifically for the Arora-Ge system of binary error LWE (coordinates of the error are chosen from $\{0, 1\}$).

Gröbner basis methods serve as a key tool for the algebraic analysis of LWE. In 2015, Albrecht et al. [2] used Gröbner bases to analyze the Arora-Ge polynomial system under the semi-regularity assumption [18, 19]. They demonstrate that this method provides an exponential speed-up over the linearization method. And for the binary error LWE (each coordinate of error is chosen from $\{0, 1\}$) with a quasi-linear sample size m (relative to the dimension n), it enables a subexponential-time solution. More recently, in Eurocrypt 2024, Steiner [32] proved that the Arora-Ge polynomial system is in generic coordinates, allowing for a proven complexity estimate of the Gröbner basis method. Moreover, the author formulated a conjecture on the degree of regularity for the system, leading to a “lowest achievable complexity estimate”. Later, Noval et al. [14] extended their methodology for the Regular Syndrome Decoding (RSD) problem to a variant of LWE, where each error coordinate lies in a fixed set of constant size. They proved that, with high probability over $\mathbb{Z}_q$, when given $m \geq \binom{n+d-1}{d}$ samples, the Arora-Ge polynomial system achieves a degree of regularity equal to d , where d is the degree of the Arora-Ge polynomial. Based on this result, their improved analysis yields a complexity that is independent of the LWE modulus q and can achieve polynomial-time complexity when a sufficient number of samples are available. Concurrently, Chen et al. [11] have extended the Arora-Ge algorithm from prime to composite modulus, thereby improving upon the original work of Arora and Ge.

The second approach is the **dual attacks**, first introduced by Micciancio and Regev [23]. Given a pair $(\mathbf{A}, \mathbf{b}) \in \mathbb{Z}_q^{m \times n} \times \mathbb{Z}_q^m$, the attacker needs to decide whether $(\mathbf{A}, \mathbf{b})$ was sampled uniformly at random or is of the form $\mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e} \bmod q$, where $(\mathbf{s}, \mathbf{e}) \in \mathbb{Z}_q^n \times \mathbb{Z}_q^m$ is a short vector. This attack exploits the structure

of the lattice

$$\Lambda_q(\mathbf{A}) = \{\mathbf{y} \in \mathbb{Z}^m \mid \exists \mathbf{x} \in \mathbb{Z}^n, \mathbf{A}\mathbf{x} = \mathbf{y} \pmod{q}\}$$

and its orthogonal lattice

$$\Lambda_q^\perp(\mathbf{A}) = \{\mathbf{x} \in \mathbb{Z}^m \mid \mathbf{A}^\top \mathbf{x} = \mathbf{0} \pmod{q}\},$$

where $\mathbb{Z}$ denotes the integer ring. The attack proceeds by computing many short vectors $\mathbf{x}$ from $\Lambda_q^\perp(\mathbf{A})$ and testing the inner products $\langle \mathbf{x}, \mathbf{b} \rangle$. For the LWE case, it tends to be biased toward small values; for the uniform case, it is uniformly distributed over $\mathbb{Z}_q$. This comes from the fact that $\langle \mathbf{x}, \mathbf{b} \rangle = \langle \mathbf{x}, \mathbf{A}\mathbf{s} + \mathbf{e} \rangle = \langle \mathbf{x}, \mathbf{e} \rangle \pmod{q}$.

In Eurocrypt 2017, this foundational idea was developed by Albrecht [1]. The author splits the matrix $\mathbf{A}$ and the secret $\mathbf{s}$ into $[\mathbf{A}_{\text{enu}} \parallel \mathbf{A}_{\text{lat}}]$ and $\begin{pmatrix} \mathbf{s}_{\text{enu}} \\ \mathbf{s}_{\text{lat}} \end{pmatrix}$, respectively. Lattice reduction algorithms are then used to find short vectors of the form $\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}$ satisfying $\mathbf{A}_{\text{lat}}^\top \mathbf{x} = \mathbf{y} \pmod{q}$. The attack proceeds by guessing $\mathbf{s}_{\text{enu}}$ and testing whether $\langle \mathbf{x}, \mathbf{b} \rangle - \langle \mathbf{y}_{\text{enu}}, \mathbf{s}_{\text{enu}} \rangle \pmod{q}$ is small, where $\mathbf{y}_{\text{enu}} = \mathbf{A}_{\text{enu}}^\top \mathbf{x} \pmod{q}$.

The evolution of dual attacks has seen successive improvements in efficiency. Espitau et al. [16] accelerated the guessing step by exploiting the recursive structure of the search space. Later, Guo and Johansson [20] used the Fast Fourier Transform (FFT) to search only a portion of the bits of $\mathbf{s}_{\text{enu}}$, which significantly reduces the search space. MATZOV [22] further replaced this search technique with “modulus-switching”, which reduces the modulus from q to a much smaller p at the expense of a controlled increase in noise. They partition the matrix into

$$[\mathbf{A}_{\text{enu}} \parallel \mathbf{A}_{\text{fft}} \parallel \mathbf{A}_{\text{lat}}] \in \mathbb{Z}_q^{m \times n_{\text{enu}}} \times \mathbb{Z}_q^{m \times n_{\text{fft}}} \times \mathbb{Z}_q^{m \times n_{\text{lat}}}$$

with $n_{\text{enu}} + n_{\text{fft}} + n_{\text{lat}} = n$, and correspondingly $\mathbf{s} = \begin{pmatrix} \mathbf{s}_{\text{enu}} \\ \mathbf{s}_{\text{fft}} \\ \mathbf{s}_{\text{lat}} \end{pmatrix}$.

The algorithm operates in two phases. First, lattice reduction algorithms are used to generate a set ζ of short vectors $\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}$ satisfying $\mathbf{A}_{\text{lat}}^\top \mathbf{x} = \mathbf{y} \pmod{q}$. These short vectors are then used to generate a new LWE sample of reduced dimension:

$$(\mathbf{a}', b') = (\mathbf{A}_{\text{fft}}^\top \mathbf{x}, \langle \mathbf{x}, \mathbf{b} - \mathbf{A}_{\text{enu}} \mathbf{s}_{\text{enu}} \rangle),$$

with secret $\mathbf{s}_{\text{fft}}$ and error $e' = \langle \mathbf{x}, \mathbf{e} \rangle + \langle \mathbf{y}, \mathbf{s}_{\text{lat}} \rangle \pmod{q}$.

To distinguish the correct guess of $\mathbf{s}_{\text{enu}}$, an evaluation function

$$E(\tilde{\mathbf{s}}_{\text{enu}}) = \max_{\mathbf{z} \in \mathbb{Z}_q^{n_{\text{fft}}}} \sum_{(\mathbf{x}, \mathbf{y}) \in \zeta} \cos\left(\frac{2\pi}{q} (\langle \mathbf{x}, \mathbf{b} - \mathbf{A}_{\text{enu}} \tilde{\mathbf{s}}_{\text{enu}} \rangle - \langle \mathbf{A}_{\text{fft}}^\top \mathbf{x}, \mathbf{z} \rangle)\right)$$

is introduced. For the correct guess, the inner expression $\langle \mathbf{x}, \mathbf{b} - \mathbf{A}_{\text{enu}} \mathbf{s}_{\text{enu}} \rangle - \langle \mathbf{A}_{\text{fft}}^\top \mathbf{x}, \mathbf{s}_{\text{fft}} \rangle$ is biased toward zero, making the sum large; for an incorrect guess

$\tilde{\mathbf{s}}_{\text{enu}} \neq \mathbf{s}_{\text{enu}}$ and any $\mathbf{z} \in \mathbb{Z}_q^{n_{\text{fft}}}$, the $\langle \mathbf{x}, \mathbf{b} - \mathbf{A}_{\text{enu}} \tilde{\mathbf{s}}_{\text{enu}} \rangle - \langle \mathbf{A}_{\text{fft}}^\top \mathbf{x}, \mathbf{z} \rangle$ is uniformly distributed over $\mathbb{Z}_q$, yielding a small sum. In practice, a threshold T is set, and a candidate $\tilde{\mathbf{s}}_{\text{enu}}$ is accepted if $E(\tilde{\mathbf{s}}_{\text{enu}}) > T$.

The evaluation can be accelerated via an FFT over $\mathbb{Z}_q^{n_{\text{fft}}}$. To further lower the FFT cost, modulus-switching reduces the modulus to a smaller $p < q$, shrinking the FFT domain to $\mathbb{Z}_p^{n_{\text{fft}}}$. The transformed evaluation function becomes

$$E'(\tilde{\mathbf{s}}_{\text{enu}}) = \max_{\mathbf{z} \in \mathbb{Z}_p^{n_{\text{fft}}}} \sum_{(\mathbf{x}, \mathbf{y}) \in \zeta} \cos\left(\frac{2\pi}{p} \left(\frac{p}{q} \langle \mathbf{x}, \mathbf{b} - \mathbf{A}_{\text{enu}} \tilde{\mathbf{s}}_{\text{enu}} \rangle - \langle \lfloor \frac{p}{q} \mathbf{A}_{\text{fft}}^\top \mathbf{x} \rfloor, \mathbf{z} \rangle\right)\right),$$

where $\lfloor \cdot \rfloor$ denotes rounding to the nearest integer. The function output still remains significantly larger for the correct $\mathbf{s}_{\text{enu}}$ than for incorrect candidates, preserving the distinguishing advantage.

Later, Ducas et al. [15] pointed out that the works of Guo and MATZOV are questionable due to an independence assumption, showing that their claimed parameters for KYBER-512 are in a contradictory regime. In response to this concern, Pouly and Shen [26] provided a provable dual attack on LWE in Eurocrypt 2024 without relying on any assumptions. Qu and Xu [27] followed in Asiacrypt 2025 with another provable dual attack that incorporated the modulus switching technique. In Crypto 2025, Carrier et al. replaced modulus switching with an efficient polar-code decoding algorithm and got rid of the questionable independence assumption in their analysis. Their work provides complexity estimates for CRYSTALS-KYBER that fall below the NIST security requirements.

On the other hand, current research on the algebraic analysis of LWE reveals its limitations in practical scenarios—particularly when the number of available samples is limited (i.e., $m = n$)—where its effectiveness falls short of that achieved by lattice attacks. In many other research areas, however, such as Arithmetization-Oriented (AO) [35, 34, 5] and Regular Syndrome Decoding (RSD) problem [7, 14], algebraic methods have been shown to be highly effective. Moreover, nearly all existing algebraic approaches for LWE build upon the foundational framework introduced by Arora and Ge, highlighting the need for further methodological advances in this direction.

1.1 Contributions

A central question arising from the progress in dual attacks is whether they can enhance algebraic analysis on LWE. Algebraically, dual lattice reduction induces a trade-off: it reduces the number of variables in the subsequent polynomial system but simultaneously amplifies the noise. This amplification presents a significant challenge for classical algebraic methods, such as directly solving the Arora-Ge system via Gröbner bases.

In this paper, we answer this question affirmatively by introducing a new algebraic algorithm that integrates dual lattice reduction into algebraic analysis. Our approach proceeds in two stages. First, we find short vectors from a lattice to generate a large set of dimension-reduced LWE samples. Second, instead of applying Gröbner basis methods to the classic Arora-Ge system, we propose

a new polynomial construction that explicitly leverages the distribution of the error and then solve the new system using a resultant-based method. Our main contributions are as follows.

1. **A New Framework for Integrating Dual Lattice Reduction into Algebraic Analysis for LWE.** We introduce a novel, two-stage framework that systematically integrates dual lattice reduction into algebraic cryptanalysis for LWE. The first stage, involves applying dual lattice reduction to transform the original LWE samples into a large number of samples with reduced dimension. The second stage solves this dimension-reduced LWE using a method different from Gröbner bases and the linearization method. The key innovation of our framework is its explicit design to leverage the *dimension-noise trade-off* induced by the first stage. Rather than treating the reduction and solving stages as independent, our framework provides a methodology to algebraically exploit this trade-off, establishing a new avenue for enhancing algebraic analysis with lattice-based methods.

2. **A New Polynomial System Construction and Resultant-Based Solving.** To overcome the limitations of applying Gröbner bases directly to the Arora-Ge system after dual lattice reduction, we introduce two interconnected algebraic advances.

First, we design a new polynomial system tailored to the dimension-reduced LWE. The classical Arora-Ge construction requires the error to be bounded within an interval smaller than the modulus q , a restriction that severely limits the achievable dimension reduction in the first stage. Our new polynomial system explicitly exploits the distribution of the amplified error after the first stage, thereby enabling more effective use of the variable reduction obtained from the dual lattice reduction. We have validated the correctness of the proposed polynomial system through small-scale experiments (see Section 3.2.2).

Second, we solve the constructed system using a resultant-based method. Although error amplification leads to a substantial growth in the total degree of the polynomials, the complexity of the resultant method is governed mainly by the number of variables and the partial degree of each individual variable. Because every coordinate of the secret $\mathbf{s}$ is bounded, we can impose a tight, explicit bound on the partial degree of each variable. This controllability constitutes a key advantage that our method actively exploits.

For CRYSTALS-KYBER, a comparison between our two-stage algorithm and the “lowest achievable complexity estimates” from prior work [32] is presented in Table 1. Although our complexity estimates remain above the NIST security levels (143/207/272 bits), they represent a significant advance over prior algebraic analysis. Compared to the “lowest achievable” Gröbner-basis estimates in [32], our approach reduces the complexity by 875/626/703 bits for KYBER-512/768/1024, respectively.

3. **An Independent Contribution of Resultant-Based Method for the Arora-Ge System.** As a separate contribution, we also show that the resultant-based method serves as a powerful algebraic solver for LWE. For

small-secret setting, applying the resultant-based method directly to the Arora-Ge system yields a provable complexity estimate that achieves an exponential speed-up over the proven bounds of Steiner [32] (Eurocrypt 2024). Concretely, for CRYSTALS-KYBER it delivers a provable complexity bound that is significantly tighter than the Gröbner bases proven estimates reported in [32] (see Table 2 for the comparison).

4. **Integrating Hints into Our Two-Stage Algorithm.** We provide a framework for incorporating various forms of side information—perfect hints, modular hints, and approximate hints—into our algorithm. This extension enhances practical applicability when partial knowledge is available and further reduces the overall complexity.

We stress that the aim of this work is not to contest the established security of LWE-based schemes, but to explore the possibility of integrating the techniques from dual attacks into algebraic analysis for LWE. Our results demonstrate the potential of this hybrid approach to refine the landscape of algebraic cryptanalysis for LWE.

1.2 Organization

The notation and foundational concepts used in this paper are introduced in Section 2. The overall framework of our two-stage algebraic algorithm is presented in Section 3: an analysis of dual lattice reduction is provided in Section 3.1; different polynomial constructions, along with solving methods, are described in Section 3.2. The independent contribution of the resultant-based method to the Arora-Ge system is demonstrated in Section 4. The incorporation of various forms of side information into the proposed algorithm is discussed in Section 5. Finally, the conclusions and some open questions are presented in Section 6.

2 Preliminaries

In this section, we present the notation and definitions used throughout the paper.

Vectors and matrices are denoted by bold lowercase and bold uppercase letters, respectively, e.g. $\mathbf{a}$ and $\mathbf{A}$. By default, all vectors in this paper are column vectors. A row vector is explicitly represented as the transpose of a column vector, e.g. $\mathbf{a}^\top$. For an integer $n \geq 1$, consider a vector $\mathbf{a} = (a_1, a_2, \dots, a_n) \in \mathbb{R}^n$. Its i -th entry is denoted by a_i for $i \in \llbracket 1, n \rrbracket$, where $\llbracket a, b \rrbracket$ denotes the set of integers from a to b , inclusive.

Let $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ denote the set of natural numbers. For a set Ω , $|\Omega|$ denotes its cardinality. The ring of integers modulo q is denoted by $\mathbb{Z}_q$, which forms a finite field if q is prime.

For vectors $\mathbf{a}, \mathbf{b} \in \mathbb{Z}_q^n$, their inner product is defined as $\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{i=1}^n a_i b_i$, which is equivalently expressed as $\mathbf{a}^\top \mathbf{b}$. The Euclidean norm of a vector $\mathbf{x} =$

$(x_1, x_2, \dots, x_n) \in \mathbb{Z}_q^n$ is defined as

$$\|\mathbf{x}\| = \sqrt{\sum_{i=1}^n \min_{j \in \mathbb{Z}} (x_i + jq)^2}.$$

Let χ be a probability distribution over $\mathbb{Z}_q$. The notation $\mathbf{a} \leftarrow \chi^n$ indicates that the vector $\mathbf{a}$ is sampled from χ^n (i.e., each entry is drawn independently from χ). The abbreviation *i.i.d.* stands for independent and identically distributed.

2.1 Lattice and Lattice Reduction

In this subsection, we will introduce some results about lattices and lattice reduction algorithms.

Definition 1. A lattice is a discrete additive subgroup of $\mathbb{R}^m$. Specifically, given n linearly independent vectors $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n \in \mathbb{R}^m$, the set

$$\Lambda = \{\mathbf{v} \mid \mathbf{v} = c_1 \mathbf{b}_1 + c_2 \mathbf{b}_2 + \dots + c_n \mathbf{b}_n, c_1, c_2, \dots, c_n \in \mathbb{Z}\}$$

is called an m -dimensional lattice. The n linearly independent vectors $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n$ is called a basis of the lattice Λ and the integer n is called the rank of Λ . A lattice Λ is called a full-rank lattice if $n = m$.

Let $\mathbf{B}$ be the $m \times n$ matrix consisting of column vectors $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n$. The determinant of the lattice Λ is defined by $\det(\Lambda) = \sqrt{\det(\mathbf{B}^\top \mathbf{B})}$, where $\mathbf{B}^\top$ is the transpose of $\mathbf{B}$. The most commonly used heuristic for lattice algorithms is the so-called Gaussian Heuristic (GH), which is heavily verified by experiments.

Definition 2 (Gaussian Heuristic [25]). Let Λ be a full-rank lattice in $\mathbb{R}^m$, and C a measurable subset of $\mathbb{R}^m$ with volume $\text{vol}(C)$. The Gaussian Heuristic “predicts” that the number of points of $\Lambda \cap C$ is roughly $\text{vol}(C) / \det(\Lambda)$.

Let $\mathbf{B} = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$ be a lattice basis of the full rank lattice Λ . Its Gram-Schmidt orthogonalized basis $\mathbf{B}^* = (\mathbf{b}_1^*, \mathbf{b}_2^*, \dots, \mathbf{b}_n^*)$ is defined as $\mathbf{b}_i^* = \mathbf{b}_i - \sum_{j=1}^{i-1} \mu_{i,j} \mathbf{b}_j^*$, where $\mu_{i,j} = \langle \mathbf{b}_i, \mathbf{b}_j^* \rangle / \|\mathbf{b}_j^*\|^2$ for $1 \leq j < i \leq n$.

Lattice reduction is the process of taking a basis for a given Λ and finding subsequent bases of Λ with shorter and closer to orthogonal vectors. The BKZ algorithm [29, 12] takes as input a lattice basis of Λ and a block size β , then outputs a BKZ- β reduced basis of Λ . Sieving algorithms is another lattice reduction algorithm, which build databases of lattice vectors, exponentially sized in the lattice dimension. In the simplest sieves, it is checked whether the sums or differences of any pair of database vectors is shorter than one of the summands or differands.

Definition 3 (Geometric Series Assumption [30]). Let $\mathbf{B}$ be a BKZ- β reduced basis with sufficiently large β , then $\|\mathbf{b}_i^*\| / \|\mathbf{b}_{i-1}^*\| \approx \delta(\beta)^{-2}$, where $\delta(\beta) = \left(\frac{\beta}{2\pi e} \cdot (\pi\beta)^{1/\beta} \right)^{\frac{1}{2(\beta-1)}}$ is the root-Hermite factor.

The geometric series assumption (GSA) is another widely used heuristic assumption in lattice-based cryptanalysis, and it is reasonably accurate for $\beta > 50$ and $\beta \ll n$ [25,12,36].

2.2 Learning with Errors

Definition 4 (LWE [28]). Let n, m and q be positive integers, let χ be a probability distribution on $\mathbb{Z}^m$ and let $\mathbf{s} \in \mathbb{Z}_q^n$ be a secret vector. A pair

$$(\mathbf{A}, \mathbf{b} = \mathbf{A} \cdot \mathbf{s} + \mathbf{e} \pmod{q}) \in \mathbb{Z}_q^{m \times n} \times \mathbb{Z}_q^m$$

is called an LWE instance, where $\mathbf{A} \in \mathbb{Z}_q^{m \times n}$ is chosen uniformly at random and $\mathbf{e} \leftarrow \chi^m$.

The LWE problem has two versions:

- **Decision-LWE:** Given a pair $(\mathbf{A}, \mathbf{b}) \in \mathbb{Z}_q^{m \times n} \times \mathbb{Z}_q^m$, which is drawn either from the LWE distribution $(\mathbf{A}, \mathbf{A} \cdot \mathbf{s} + \mathbf{e} \pmod{q})$ or from the uniform distribution on $\mathbb{Z}_q^{m \times n} \times \mathbb{Z}_q^m$, determine which distribution it comes from.
- **Search-LWE:** Given an LWE instance $(\mathbf{A}, \mathbf{b} = \mathbf{A} \cdot \mathbf{s} + \mathbf{e} \pmod{q}) \in \mathbb{Z}_q^{m \times n} \times \mathbb{Z}_q^m$, recover the secret vector $\mathbf{s}$.

In most practical LWE-based cryptosystems, the secret and error vectors are both small, where their coordinates are chosen from a small, fixed interval. **We adopt and consistently rely on this common setting throughout this paper.**

In particular, CRYSTALS-KYBER employs a centered binomial distribution for both the secret and error coordinates, which is defined as follows.

Definition 5 (Centered Binomial Distribution). The centered binomial distribution B_η with parameter $\eta \in \llbracket 0, \frac{q-1}{2} \rrbracket$ is defined as

$$B_\eta \sim \sum_{i=1}^{\eta} (X_i - Y_i),$$

where X_i and Y_i are i.i.d. uniform over $\{0, 1\}$.

It is well-known that B_η has mean 0 and standard deviation $\sigma = \sqrt{\frac{\eta}{2}}$. The probability that $B_\eta = i$ is given by $\mathbb{P}(B_\eta = i) = 2^{-2\eta} \binom{2\eta}{\eta+i}$ for $i \in \llbracket -\eta, \eta \rrbracket$. In the KYBER schemes, $\eta = 3$ for KYBER-512 and $\eta = 2$ for KYBER-768/1024.

2.3 Modular Gaussian Distribution

Let $\sigma > 0$. For all $x \in \mathbb{R}$, the density of the centered Gaussian distribution with standard deviation σ is defined as $\rho_\sigma(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x^2}{2\sigma^2}\right)$.

Definition 6 (Modular Gaussian Distribution). *The Modular Gaussian distribution mod q is a probability distribution over $\mathbb{Z}_q$, defined by the probability mass function*

$$\rho_{q,\sigma}(k) = \frac{1}{C_{q,\sigma}} \sum_{x \in k+q\mathbb{Z}} \exp\left(-\frac{x^2}{2\sigma^2}\right),$$

where $C_{q,\sigma} = \sum_{k \in \mathbb{Z}} \exp\left(-\frac{k^2}{2\sigma^2}\right)$ is a normalization constant.

For a random variable X following a normal distribution with mean 0 and variance σ^2 , one can utilize the well-known Mills' inequality

$$\mathbb{P}[|X| > t\sigma] \leq \frac{2}{t\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right). \quad (1)$$

In Section 3.1, we utilize Eq. (1) to estimate the bound of the error's coordinates after dual lattice reduction.

2.4 Gröbner Bases

In this subsection, we give a brief introduction to the theory of Gröbner bases. For more details, refer to [13].

Let $\mathbb{K}$ be a field and $\mathbb{K}[x_1, x_2, \dots, x_n]$ the polynomial ring over $\mathbb{K}$ with indeterminates $x_1, x_2, \dots, x_n$. An expression of the form $x_1^{k_1} x_2^{k_2} \dots x_n^{k_n}$ with each $k_i \in \mathbb{N}$ is called a *term*. The set of all such terms is written as $T(x_1, x_2, \dots, x_n)$. A *monomial* is a term multiplied by a nonzero coefficient $a \in \mathbb{K}$, i.e., $ax_1^{k_1} \dots x_n^{k_n}$.

A *term order* on $T(x_1, x_2, \dots, x_n)$ is a linear order $\preceq$ satisfying:

- For every term $X \in T(x_1, x_2, \dots, x_n)$, $1 \preceq X$.
- For all terms $Y, X_1, X_2 \in T(x_1, x_2, \dots, x_n)$, $X_1 \preceq X_2$ implies $YX_1 \preceq YX_2$.

For a polynomial $f \in \mathbb{K}[x_1, \dots, x_n]$, its *leading term* with respect to $\preceq$, denoted by $\text{LT}_{\preceq}(f)$, is the largest term appearing in f . For an ideal $I \subseteq \mathbb{K}[x_1, \dots, x_n]$, its *initial ideal* is defined as

$$\text{in}_{\preceq}(I) = \langle \text{LT}_{\preceq}(f) \mid f \in I \rangle.$$

Definition 7 (Gröbner basis). *Let $I \subseteq \mathbb{K}[x_1, \dots, x_n]$ be an ideal and $\preceq$ a term order. A finite subset $\mathcal{G} = \{g_1, \dots, g_m\} \subseteq I$ is said to be a Gröbner basis of I with respect to $\preceq$ if $0 \notin \mathcal{G}$ and*

$$\text{in}_{\preceq}(I) = \langle \text{LT}_{\preceq}(g_1), \dots, \text{LT}_{\preceq}(g_m) \rangle.$$

For a given polynomial system $\mathcal{F} = \{f_1, \dots, f_m\} \subseteq \mathbb{K}[x_1, \dots, x_n]$ and a given term order $\preceq$, the computation of a Gröbner basis of the ideal $\langle \mathcal{F} \rangle$ is often performed via linear algebra on large structured matrices, known as Macaulay matrices. Recall that a polynomial is said to be *homogeneous* if all its terms have the same degree. For a positive integer d , a *homogeneous Macaulay matrix* M_d is

constructed as follows: its columns are indexed by all terms of degree d , ordered from left to right with respect to $\preceq$. The rows are indexed by all polynomials of the form $s \cdot f_i$, where $s \in T(x_1, x_2, \dots, x_n)$ is a term satisfying $\deg(s \cdot f_i) = d$. The entry of a row $s \cdot f_i$ in column t is the coefficient of $s \cdot f_i$ at the term t . For an inhomogeneous system, one considers instead the matrix $M_{\leq d}$, whose columns are indexed by all terms of degree less than or equal to d and whose rows correspond to all $s \cdot f_i$ with $\deg(s \cdot f_i) \leq d$.

A key parameter governing the complexity of computing a Gröbner basis is the solving degree.

Definition 8 (Solving degree, [8]). *Let $\mathcal{F} = \{f_1, \dots, f_m\} \subseteq \mathbb{K}[x_1, \dots, x_n]$ and let $\preceq$ be a term order. The solving degree of $\mathcal{F}$, denoted by $\text{sd}_{\preceq}(\mathcal{F})$, is the least integer d such that Gaussian elimination on the Macaulay matrix $M_{\leq d}$ produces a Gröbner basis of $\langle \mathcal{F} \rangle$ with respect to $\preceq$.*

In this paper, we employ the solving degree concerning the degree reverse lex (DRL) order.

Definition 9 (Degree Reverse Lex Order [13]). *Let $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)$, $\beta = (\beta_1, \beta_2, \dots, \beta_n) \in \mathbb{N}^n$. We say*

$$x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n} \preceq_{\text{DRL}} x_1^{\beta_1} x_2^{\beta_2} \cdots x_n^{\beta_n},$$

if $\sum_{i=1}^n \alpha_i < \sum_{i=1}^n \beta_i$, or $\sum_{i=1}^n \alpha_i = \sum_{i=1}^n \beta_i$ and the right-most nonzero entry of $\alpha - \beta$ is positive.

For a system $\mathcal{F}$ of m polynomials in n variables with solving degree $\text{sd} = \text{sd}_{\text{DRL}}(\mathcal{F})$, the cost of computing its Gröbner basis with respect to the DRL order is bounded by

$$O\left(m \cdot \text{sd} \cdot \binom{n + \text{sd} - 1}{\text{sd}}^\omega\right), \quad (2)$$

where $2 \leq \omega < 3$ is the linear algebra exponent¹.

Denote the highest component of f as f^{top} . For a given finite set of polynomials $\mathcal{F} = \{f_1, \dots, f_m\} \subseteq \mathbb{K}[x_1, x_2, \dots, x_n]$, define $\mathcal{F}^{\text{top}} = \{f_1^{\text{top}}, \dots, f_m^{\text{top}}\}$.

Another key parameter that characterizes the complexity of solving a polynomial system is the degree of regularity.

Definition 10 (Degree of Regularity, [17, Definition 4]). *Let $\mathbb{K}$ be a field and $\mathcal{F} \subseteq \mathbb{K}[x_1, \dots, x_n]$. Assume that $\langle \mathcal{F}^{\text{top}} \rangle_d = P_d$ for some integer $d \geq 0$, where $\langle \mathcal{F}^{\text{top}} \rangle_d$ and P_d are the sets of homogeneous polynomials of degree d in the ideal $\langle \mathcal{F}^{\text{top}} \rangle$ and $\mathbb{K}[x_1, \dots, x_n]$, respectively. The degree of regularity is defined as*

$$d_{\text{reg}}(\mathcal{F}) = \min \{d \geq 0 \mid \langle \mathcal{F}^{\text{top}} \rangle_d = P_d\}.$$

In [32], the author employs a conjecture that sets a bound on the solving degree in relation to the degree of regularity.

¹ We set $\omega = 2$ to maintain consistency with the complexity analysis used in [32].

Conjecture 1 ([32, Conjecture 25]). Let q be a prime number and let $\mathcal{F}_{\text{AG}} \subseteq \mathbb{Z}_q[x_1, \dots, x_n]$ be the Arora-Ge polynomial system. For the small secret LWE, if each coordinate of the error is chosen from $\llbracket -T, T \rrbracket$, then

$$\text{sd}_{\text{DRL}}(\mathcal{F}_{\text{AG}}) \leq d_{\text{reg}}(\mathcal{F}_{\text{AG}}) + 2T.$$

Consequently, the complexity of computing a Gröbner basis of $\langle \mathcal{F}_{\text{AG}} \rangle$ with respect to the DRL order can be estimated by substituting the bound $d_{\text{reg}}(\mathcal{F}_{\text{AG}}) + 2T$ for solving degree in the Eq. (2).

2.5 Resultant

The resultant is a powerful tool for solving polynomial system. Recent studies on Arithmetization-Oriented (AO) primitives have demonstrated that, in certain settings, resultant-based solvers can outperform traditional Gröbner basis methods [5, 35, 34]. Motivated by these findings, we adopt a resultant-based method for solving the new polynomial system in this paper.

Definition 11 (Resultant). Let q be a prime number and let $f(\mathbf{x}, y), g(\mathbf{x}, y) \in \mathbb{Z}_q[\mathbf{x}, y]$ with $\mathbf{x} = (x_1, \dots, x_s)$. The resultant of $f(\mathbf{x}, y)$ and $g(\mathbf{x}, y)$ with respect to the indeterminate y , denoted by $R(f, g, y)$, is defined as the determinant of the Sylvester matrix of $f(\mathbf{x}, y)$ and $g(\mathbf{x}, y)$ when considered as polynomials in the single indeterminate y . That is, if $f(\mathbf{x}, y) = \sum_{i=0}^u f_i y^i$ and $g(\mathbf{x}, y) = \sum_{i=0}^v g_i y^i$, where $f_i, g_i \in \mathbb{Z}_q[\mathbf{x}]$, then

$$R(f, g, y) = \begin{vmatrix} f_u & f_{u-1} & \cdots & f_0 & & & \\ & f_u & f_{u-1} & \cdots & f_0 & & \\ & & \ddots & \ddots & & \ddots & \\ & & & f_u & f_{u-1} & \cdots & f_0 \\ g_v & g_{v-1} & \cdots & \cdots & g_0 & & \\ & g_v & g_{v-1} & \cdots & \cdots & g_0 & \\ & & \ddots & \ddots & \ddots & \ddots & \\ & & & g_v & g_{v-1} & \cdots & g_0 \end{vmatrix}$$

It is well-known that the resultant of two polynomials is non-zero if and only if they have no common factors other than non-zero constants. In this case, the resultant yields a new polynomial $h(\mathbf{x})$, such that if $(\mathbf{x}_0, y_0)$ is a common root of both $f(\mathbf{x}, y)$ and $g(\mathbf{x}, y)$, then $h(\mathbf{x}_0) = 0$. In this way, we can remove one variable from two polynomials while retaining information about the roots of the original polynomials. For a system of ℓ polynomials in s variables, we can repeatedly compute resultants of the polynomials until we obtain a univariate polynomial. Solving for the roots of the univariate polynomial and repeatedly substituting them back, we can derive the roots that the polynomials have in common. Lemma 1 shows the cost of multivariate polynomial multiplication over $\mathbb{Z}_q$.

Lemma 1 (Complexity of multivariate polynomial multiplication, [5]). *Let $f, g \in \mathbb{Z}_q[x_1, \dots, x_n]$ be two polynomials with $\deg_{x_i}(f) = \alpha_i$ and $\deg_{x_i}(g) = \beta_i$ for $i = 1, \dots, n$, where $\deg_{x_i}(f)$ is the partial degree of f with respect to x_i . Using the Kronecker substitution technique [24], their product $h = f \cdot g$ can be computed with*

$$\mathcal{M}\left(\prod_{i=1}^n (\alpha_i + \beta_i + 1)\right)$$

operations in $\mathbb{Z}_q$, where $\mathcal{M}(d) = \mathcal{O}(d \log d \log \log d)$. The resulting polynomial satisfies $\deg_{x_i}(h) = \alpha_i + \beta_i$ and has at most $\prod_{i=1}^n (\alpha_i + \beta_i + 1)$ non-zero coefficients.

Lemma 2 provides an estimation for finding the roots of a univariate polynomial over $\mathbb{Z}_q$.

Lemma 2 (Complexity of univariate polynomial root finding, [6, Section 3.1]). *Let $f(x) \in \mathbb{Z}_q[x]$ be a univariate polynomial of degree d . Using an FFT-based algorithm, all roots of $f(x)$ in $\mathbb{Z}_q$ can be found in*

$$\mathcal{O}(d \log(d) (\log(d) + \log(q)) \log(\log(d)))$$

operations over $\mathbb{Z}_q$.

3 A Two-Stage Algebraic Analysis Framework for LWE

Recent advances in dual attacks have demonstrated their effectiveness for LWE. In contrast, existing algebraic attacks on LWE are far from satisfactory. This gap motivates us to investigate whether integrating dual lattice reduction into algebraic analysis can yield improvements.

In this section, we give a novel, two-stage hybrid framework that systematically integrates dual lattice reduction into algebraic cryptanalysis for LWE. In the first stage, short vectors are extracted from a lattice to produce a large set of dimension-reduced LWE samples. This transformation offers an algebraic advantage: it significantly decreases the number of unknown variables while simultaneously providing abundant samples. However, it also introduces a drawback: the noise is amplified, increasing the magnitude of the error. Consequently, applying the classic Arora-Ge method directly to those transformed samples is ineffective since it merely enumerates all possible bounded error values without exploiting their distribution, which naturally requires that the size of the error range must be smaller than the modulus q . Moreover, the high total degree of the resulting polynomial system also makes Gröbner basis computations impractical. To overcome these limitations, in the second stage we propose a novel polynomial construction that explicitly exploits the distribution of the amplified error—in fact, the probability that it equals zero. This is achieved by selecting a random set of K distinct coordinates and multiplying them together (the details are provided in Section 3.2.2). This approach allows us to build the polynomial

system without requiring that the size of error range be smaller than the modulus q , which permits a larger dimension in the dual lattice reduction stage. This new system is solved using a resultant-based method, whose complexity is governed mainly by the number of variables and the partial degree in each variable—the latter being a parameter that is kept small and tightly bounded in our construction.

The framework of our proposed algorithm is outlined in Algorithm 1.

Algorithm 1 A New Algebraic Analysis Framework for LWE

Input: An LWE instance $(\mathbf{A}, \mathbf{b}) \in \mathbb{Z}_q^{m \times n} \times \mathbb{Z}_q^m$, parameters $n_{\text{lat}}, n_{\text{solve}} = n - n_{\text{lat}}, \beta_{\text{sieve}}, \beta_{\text{bkz}}$.

Output: Secret vector $\mathbf{s}$.

- 1: Partition $\mathbf{A}$ column-wise into $[\mathbf{A}_{\text{solve}} \parallel \mathbf{A}_{\text{lat}}]$, where $\mathbf{A}_{\text{solve}} \in \mathbb{Z}_q^{m \times n_{\text{solve}}}, \mathbf{A}_{\text{lat}} \in \mathbb{Z}_q^{m \times n_{\text{lat}}}$, correspondingly split $\mathbf{s} = \begin{pmatrix} \mathbf{s}_{\text{solve}} \\ \mathbf{s}_{\text{lat}} \end{pmatrix}$.
- 2: $\zeta \leftarrow \text{ShortVectorSampling}(\mathbf{A}_{\text{lat}}, \beta_{\text{bkz}}, \beta_{\text{sieve}})$. $\triangleright |\zeta| = N \approx (4/3)^{\beta_{\text{sieve}}/2}$
- 3: $\Omega \leftarrow \text{GenerateNewSamples}(\zeta, \mathbf{A}_{\text{solve}}, \mathbf{b})$.
- 4: $\mathbf{s}_{\text{solve}} \leftarrow \text{PolynomialSystemSolving}(\Omega)$.
- 5: $\mathbf{s}_{\text{lat}} \leftarrow \text{SolveSubLWE}(\mathbf{A}_{\text{lat}}, \mathbf{b}, \mathbf{s}_{\text{solve}})$. $\triangleright$ Recover remaining secret components

Return Secret vector $\mathbf{s} = (\mathbf{s}_{\text{solve}}^\top \parallel \mathbf{s}_{\text{lat}}^\top)^\top$.

Steps 1-3 constitute the first stage, dual lattice reduction, whose detailed analysis is provided in Section 3.1.

Step 4 forms the second stage, polynomial construction and solving, which is described in Section 3.2.

Once $\mathbf{s}_{\text{solve}}$ is recovered, the remaining secret coordinates can be obtained by solving a smaller sub-LWE instance (Step 5). We do not prescribe a specific subroutine for this final step, as its cost is dominated by the preceding stages of Algorithm 1.

In this section, we consider LWE with a small secret and a small error, wherein all coordinates are chosen from a fixed small interval $\llbracket -T, T \rrbracket$. This is the setting for most practical LWE-based schemes.

3.1 The First Stage: Constructing Dimension-reduced LWE via Dual Lattice Reduction

The first stage follows the methodology from dual attacks [20, 22, 10]. Specifically, we first apply lattice reduction algorithms to the lattice generated by the $r \times r$ matrix

$$\mathbf{B} = \begin{bmatrix} \mathbf{I}_m & \mathbf{0} \\ \mathbf{A}_{\text{lat}}^\top & q\mathbf{I}_{n_{\text{lat}}} \end{bmatrix},$$

where $r = m + n_{\text{lat}}$. This yields N short vectors of the form $\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}$ satisfying $\mathbf{A}_{\text{lat}}^\top \mathbf{x} = \mathbf{y} \pmod{q}$ (See Algorithm 2 for details).

Algorithm 2 Short Vectors Sampling [20]

Input: A basis $\mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2 \dots, \mathbf{b}_r]$ of a lattice, and integers $2 \leq \beta_{\text{bkz}}, \beta_{\text{sieve}} \leq r$.

Output: A set ζ of $N_{\text{sieve}}(\beta_{\text{sieve}})$ vectors from the lattice.

- 1: Run BKZ- β_{bkz} on $\mathbf{B}$ to obtain a reduced basis $\mathbf{b}'_1, \mathbf{b}'_2 \dots, \mathbf{b}'_r$.
- 2: Run a sieve in dimension β_{sieve} on the sublattice spanned by $\mathbf{b}'_1, \dots, \mathbf{b}'_{\beta_{\text{sieve}}}$ and collect the output into a set ζ .

Return: ζ .

The expected number of short vectors given by Algorithm 2 is

$$N_{\text{sieve}}(\beta_{\text{sieve}}) \approx \left(\sqrt{\frac{4}{3}} \right)^{\beta_{\text{sieve}}}.$$

Each such vector is then used to construct a new dimension-reduced LWE sample $(\mathbf{a}', b')$ satisfying

$$b' = \langle \mathbf{a}', \mathbf{s}_{\text{solve}} \rangle + e' \pmod{q},$$

where

$$\mathbf{a}' = \mathbf{A}_{\text{solve}}^\top \mathbf{x}, \quad b' = \langle \mathbf{b}, \mathbf{x} \rangle, \quad e' = \langle \mathbf{x}, \mathbf{e} \rangle + \langle \mathbf{y}, \mathbf{s}_{\text{lat}} \rangle.$$

The expected length of the vectors produced by Algorithm 2 can be estimated by the following Lemma 3.

Lemma 3 (Length of vectors from Algorithm 2, [10, Lemma 2.8]). *Let Λ be the lattice generated by the $r \times r$ matrix $\mathbf{B}$ in Algorithm 2, and let β_{bkz} and β_{sieve} be the blocksizes for BKZ and sieve, respectively. If the Gaussian Heuristic (GH) and the Geometric Series Assumption (GSA) hold for Λ , then Algorithm 2 outputs $N_{\text{sieve}}(\beta_{\text{sieve}}) \approx \left(\sqrt{\frac{4}{3}} \right)^{\beta_{\text{sieve}}}$ vectors of expected length ℓ given by*

$$\ell = \det(\Lambda)^{1/r} \cdot \sqrt{\frac{4}{3}} \cdot \delta(\beta_{\text{sieve}})^{\beta_{\text{sieve}}-1} \cdot \delta(\beta_{\text{bkz}})^{r-\beta_{\text{sieve}}},$$

where $\delta(\beta) = \left(\frac{\beta}{2\pi e} \cdot (\pi\beta)^{1/\beta} \right)^{\frac{1}{2(\beta-1)}}$ is the root-Hermite factor.

To finish this subsection, we calculate the probability distribution of e' in Section 3.1.1, and then analyze the complexity of the first stage in Section 3.1.2.

3.1.1 The Probability Distribution of e'

We first invoke the following assumption regarding the distribution of the sampled short vectors.

Assumption 1 ([12,16,22]) *Let ℓ be the expected length of vectors returned by Algorithm 2. Then the coordinates of the returned vectors have approximately independent Gaussian distributions with mean 0 and standard deviation $\ell/\sqrt{r}$, where r is the dimension of the matrix $\mathbf{B}$.*

Now we give the probability distribution of e' .

Lemma 4 (Probability distribution of e'). *Under Assumption 1, the quantity $e' = \langle \mathbf{x}, \mathbf{e} \rangle + \langle \mathbf{y}, \mathbf{s}_{\text{lat}} \rangle$ follows a modular Gaussian distribution with mean 0 and variance ²*

$$\sigma^2 \approx (\|\mathbf{e}\|^2 + \|\mathbf{s}_{\text{lat}}\|^2) \cdot \frac{\ell^2}{m + n_{\text{lat}}}.$$

Proof. By Assumption 1, the coordinates of $\mathbf{x}$ and $\mathbf{y}$ are approximately independent Gaussians with mean 0 and variance ℓ^2/r , where $r = m + n_{\text{lat}}$. Hence, $\langle \mathbf{x}, \mathbf{e} \rangle$ and $\langle \mathbf{y}, \mathbf{s}_{\text{lat}} \rangle$ are multiple sums of independent Gaussian variables and are themselves modular Gaussian. Therefore, the mean of $\langle \mathbf{x}, \mathbf{e} \rangle + \langle \mathbf{y}, \mathbf{s}_{\text{lat}} \rangle$ is 0 and the variance

$$\begin{aligned} \sigma^2 &\approx \text{Var}\left(\sum_{i=1}^m x_i e_i + \sum_{j=1}^{n_{\text{lat}}} y_j (s_{\text{lat}})_j\right) \\ &= \mathbb{E}\left(\sum_{i=1}^m x_i^2 e_i^2 + \sum_{j=1}^{n_{\text{lat}}} y_j^2 (s_{\text{lat}})_j^2\right) \\ &= (\|\mathbf{e}\|^2 + \|\mathbf{s}_{\text{lat}}\|^2) \cdot \frac{\ell^2}{m + n_{\text{lat}}} \end{aligned}$$

which is the stated formula. $\square$

From the definition of the modular Gaussian distribution (see Definition 6), the probability that $e' = k \pmod q$ can be given by

$$\mathbb{P}(e' = k) = \frac{1}{C_{q,\sigma}} \sum_{x \in k+q\mathbb{Z}} \exp\left(-\frac{x^2}{2\sigma^2}\right), \quad (3)$$

where $C_{q,\sigma} = \sum_{k \in \mathbb{Z}} \exp(-\frac{k^2}{2\sigma^2})$ is the normalization constant. In practice, infinite sums can be approximated by truncating them to a sufficiently large finite range. In Section 3.2.2, we will propose a polynomial construction that uses $\mathbb{P}(e' = k)$.

3.1.2 Complexity of the First Stage

The complexity of the first stage comprises two parts:

- **Short vector sampling**, the cost of which is stated in Lemma 5.
- **Generation of new LWE samples**, which costs $\mathcal{O}(\sqrt{\frac{4}{3}}^{\beta_{\text{sieve}}})$.

Lemma 5. *The complexity of Algorithm 2 is*

$$T_{\text{sample}} = T_{\text{BKZ}} + T_{\text{sieve}},$$

where T_{BKZ} and T_{sieve} denote the costs of the BKZ and the sieve, respectively.

² In practice, verifying the independence among the coordinates of the vectors returned by Algorithm 2 is impractical. Therefore, in our subsequent analysis, we adopt the variance estimate $\sigma^2 = (\|\mathbf{e}\|^2 + \|\mathbf{s}_{\text{lat}}\|^2) \cdot \frac{\ell^2}{m+n_{\text{lat}}}$, as employed in [22].

In fact, from the subsequent analysis, the primary complexity is not occupied by the first stage, so we do not analyze the differences in model complexity for lattice reduction algorithms. Consequently, we adopt the “core-SVP” cost model³ (denoted as the **C0** model in [10, Section 5]) for Algorithm 2. This model assumes that a single call to an SVP oracle suffices to reduce a lattice basis and that all lower-order terms in the exponent are negligible. Within this framework, the complexity of Algorithm 2 is given by $\mathcal{O}(2^{0.292\beta})$, where $\beta = \beta_{\text{bkz}} = \beta_{\text{sieve}}$.

3.2 The Second Stage: Polynomial Construction and Solving

After the first stage, we now have $N = N_{\text{sieve}}(\beta_{\text{sieve}})$ new dimension-reduced LWE samples

$$(\mathbf{a}'_i, b'_i = \langle \mathbf{a}'_i, \mathbf{s}_{\text{solve}} \rangle + e'_i \pmod{q}), i = 1, 2, \dots, N,$$

where each coordinate of $\mathbf{s}_{\text{solve}}$ is chosen from the integer interval $\llbracket -T, T \rrbracket$ and each e'_i is bounded (with high probability) within $\llbracket -T', T' \rrbracket$ with $T' = \lceil t\sigma \rceil$. Here, the parameter t is taken to be sufficiently large so that, with high probability, each e'_i lies within this bounded interval; and

$$\sigma = \sqrt{(\|\mathbf{e}\|^2 + \|\mathbf{s}_{\text{lat}}\|^2) \cdot \frac{\ell^2}{m + n_{\text{lat}}}},$$

where ℓ is the expected length of the short vectors produced by Algorithm 2, given by Lemma 3. For convenience, $\mathbf{e}' = (e'_1, e'_2, \dots, e'_N)$ is called the amplified error vector.

The second stage constructs a polynomial system from the dimension-reduced samples and solves it using two different algebraic approaches. In Section 3.2.1, we construct the polynomial system via the Arora-Ge method (hereafter referred to as the Arora-Ge system) and solve it with Gröbner bases; the corresponding complexity estimate follows the framework established in [32]. In Section 3.2.2, we propose a new polynomial system that exploits the error distribution and solve it using a resultant-based method. The results demonstrate that our new construction, combined with the resultant-based method, yields a more effective synergy with the preceding stage.

3.2.1 Construction of the Arora-Ge System and Solving by the Gröbner Basis Method

Although the first stage can produce substantially dimension-reduced samples, directly applying the Gröbner basis method to solve the resulting Arora-Ge system proves ineffective.

More precisely, by applying the polynomial construction introduced by Arora and Ge [4], the Arora-Ge system is defined as

$$\mathcal{F}_{\text{AG}} \triangleq \{f_1(b'_1 - \langle \mathbf{a}'_1, \mathbf{s}_{\text{solve}} \rangle), \dots, f_N(b'_N - \langle \mathbf{a}'_N, \mathbf{s}_{\text{solve}} \rangle)\},$$

³ For a more precise estimate, one can employ the “lattice estimator” of [3].

where each polynomial f_i is obtained by substituting the linear form $b'_i - \langle \mathbf{a}'_i, \mathbf{s}_{\text{solve}} \rangle$ into the univariate polynomial $f(X) = X \prod_{j=1}^{\lceil t\sigma \rceil} (X + j)(X - j)$, which has a degree of $D = 2\lceil t\sigma \rceil + 1$.

Among the three distinct complexity-estimation methods presented in [32], we employ the “lowest achievable complexity estimate” for our analysis. This choice provides a fairer and more demanding comparison. Theorem 1 states the complexity of solving $\mathcal{F}_{\text{AG}}$ using the Gröbner basis method under this “lowest estimate”.

Theorem 1. *Let $q \geq 2$, $m = n \geq 2$ be LWE parameters, with the secret $\mathbf{s}$ and error $\mathbf{e}$ having coordinates chosen from $\llbracket -T, T \rrbracket$. After performing dual lattice reduction with parameters n_{lat} , $n_{\text{solve}} = n - n_{\text{lat}}$, β_{bkz} , and β_{sieve} , let $\mathcal{F}_{\text{AG}}$ denote the Arora-Ge system constructed from the resulting N dimension-reduced samples. With a success probability of at least $1 - \epsilon$, one can compute a Gröbner basis for the ideal $\langle \mathcal{F}_{\text{AG}} \rangle$ with respect to the DRL order and its time complexity is ⁴*

$$O\left(N \cdot \text{sd} \cdot \binom{n_{\text{solve}} + \text{sd} - 1}{\text{sd}}^{\omega}\right),$$

where $N = (\sqrt{4/3})^{\beta_{\text{sieve}}}$, and $\text{sd} = d_{\min} + 4\lceil t\sigma \rceil + 1$. Here, $d_{\min}$ is the smallest integer d satisfying the inequality

$$N \cdot \binom{n_{\text{solve}} + d - 1}{d} \geq \binom{n_{\text{solve}} + D + d - 1}{D + d}, \quad (4)$$

with $D = 2\lceil t\sigma \rceil + 1$. The standard deviation σ of the amplified error is given by

$$\sigma = \sqrt{(\|\mathbf{e}\|^2 + \|\mathbf{s}_{\text{lat}}\|^2) \cdot \frac{\ell^2}{m + n_{\text{lat}}}},$$

in which $\ell = \det(\Lambda)^{1/r} \cdot \sqrt{4/3} \cdot \delta(\beta_{\text{sieve}})^{\beta_{\text{sieve}} - 1} \cdot \delta(\beta_{\text{bkz}})^{r - \beta_{\text{sieve}}}$ with $r = m + n_{\text{lat}}$. The parameter t is chosen such that the success probability

$$p_{\text{success}} \geq 1 - N \cdot \frac{2}{t\sqrt{2\pi}} \exp(-t^2/2) = 1 - \epsilon.$$

Proof. It immediately follows from Eq. (1) that t is chosen such that every coordinate of the amplified error vector $\mathbf{e}' = (e'_1, e'_2, \dots, e'_N)$ falls within an interval $\llbracket -\lceil t\sigma \rceil, \lceil t\sigma \rceil \rrbracket$ with a probability of at least $1 - \epsilon$. Consequently, the degree of the polynomials in $\mathcal{F}_{\text{AG}}$ is $D = 2\lceil t\sigma \rceil + 1$, and so the degree of regularity $d_{\text{reg}}(\mathcal{F}_{\text{AG}}) = d_{\min} + D$. Finally, applying Conjecture 1 provides the upper bound of the solving degree $\text{sd}_{\text{DRL}}(\mathcal{F}_{\text{AG}}) \leq d_{\text{reg}}(\mathcal{F}_{\text{AG}}) + 2\lceil t\sigma \rceil$. Substituting this bound into Eq. (2) completes the proof. $\square$

⁴ To recover the true secret from this basis, an additional step—such as the FGLM algorithm or the Gröbner Walk—is required to convert the DRL basis into a lexicographic one. In this paper, the complexity of this order conversion is omitted.

We estimate the time complexity for CRYSTALS-KYBER using Theorem 1. The results, presented in Figure 1, show that the cost of the first stage is negligible. To minimize the adverse effect of the dual lattice reduction stage on the subsequent stage, we set the blocksize parameter β_{bkz} and β_{sieve} to its maximal feasible value, i.e., $\beta_{\text{bkz}} = \beta_{\text{sieve}} = m + n_{\text{lat}}$. And we fix the overall success probability at 0.99. Under this setting, the best complexity estimation—obtained when $n_{\text{lat}} = 1$ —remains prohibitively high and substantially exceeds the estimates from [32], which are 1584, 1588, and 2014 for KYBER-512/768/1024, respectively. The primary reason is that the rapid growth of the polynomial degree D becomes the dominant factor driving the overall complexity. Consequently, constructing the Arora-Ge system and solving it via Gröbner bases in the second stage proves to be an ineffective strategy.

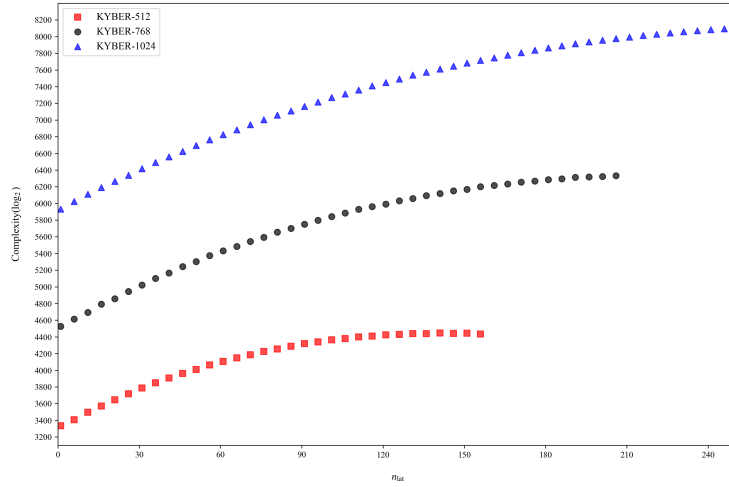


Fig. 1. Complexity estimation with the increase of n_{lat} for CRYSTALS-KYBER. The horizontal axis represents n_{lat} , and the vertical axis represents the complexity ($\log_2$). The parameter $\beta_{\text{bkz}} = \beta_{\text{sieve}} = r$, where r is the dimension of $\mathbf{B}$ (see Algorithm 2). The growth of n_{lat} is restricted by the condition that $T = \lceil t\sigma \rceil < q/2$.

3.2.2 Construction of a New Polynomial System and Solving by the Resultant-based Method

In Section 3.2.1, we have shown that constructing the Arora-Ge system and then solving it with the Gröbner basis method leads to prohibitively high complexity in the second stage. An analysis of the underlying causes reveals that the Arora-Ge system merely enumerates all possible values of the amplified error without exploiting their actual distribution; moreover, the rapid growth of the polynomial degree negatively affects the overall complexity. To exploit this specific distributional feature, we design a new polynomial system that explic-

itly incorporates the zero-error probability and solve it using a resultant-based method.

Construction of a New Polynomial System. Motivated by the observation that the amplified error follows a modular Gaussian distribution with a mean of zero, we construct a polynomial system that focuses on those high-probability values rather than enumerating all possibilities. This approach completely removes the requirement that the size of the error range must be smaller than the modulus q . However, it also reduces the probability that the resulting equations hold for the true secret. To compensate, we leverage the large number of new LWE samples produced in the first stage and form each polynomial from multiple samples, thereby increasing the success probability.

We first present the general form of this new polynomial system. For an integer $0 \leq R \leq T'$, we define the polynomial

$$F = \prod_{i=1}^K X_i \cdot \prod_{i=1}^K (X_i + 1)(X_i - 1) \cdots \prod_{i=1}^K (X_i + R)(X_i - R). \quad (5)$$

Notably, setting $K = 1$ and $R = T'$ reduces Eq. (5) exactly to the original Arora-Ge polynomial.

In practice, we adopt a streamlined construction that directly exploits the highest probability, which is the zero-probability. Specifically, we consider the polynomial

$$F = \prod_{i=1}^K X_i.$$

To obtain a concrete multivariate polynomial, we randomly select K distinct indices $\{j_1, j_2, \dots, j_K\}$ from the sample set Ω and replace each variable X_i with the linear form $b'_{j_i} - \langle \mathbf{a}'_{j_i}, \mathbf{s}_{\text{solve}} \rangle$ for $i = 1, 2, \dots, K$. This yields

$$F = \prod_{i=1}^K (b'_{j_i} - \langle \mathbf{a}'_{j_i}, \mathbf{s}_{\text{solve}} \rangle).$$

By selecting W such polynomials $F_1, F_2, \dots, F_W$ from disjoint index sets (meaning that all selected samples are distinct), we construct the polynomial system

$$\mathcal{F}_{\text{new}} \triangleq \{F_1, F_2, \dots, F_W\}. \quad (6)$$

Two fundamental questions naturally arise: (1) What is the probability that the true secret satisfies the system $\mathcal{F}_{\text{new}}$? (2) What is the expected number of solutions to this system? Let $\mathbf{s}_{\text{solve}}$ be the true secret vector and $\tilde{\mathbf{s}} \neq \mathbf{s}_{\text{solve}}$ an arbitrary incorrect candidate.

First, given the probability $p_0 = \mathbb{P}(e'_i = 0)$ calculated by Eq. (3), the probability that the true secret is a solution for $\mathcal{F}_{\text{new}}$ can be estimated as

$$p = (1 - (1 - p_0)^K)^W. \quad (7)$$

Second, the expected number of solutions is characterized by the following lemma.

Lemma 6. *Let $\mathcal{F}_{new} = \{F_1, F_2, \dots, F_W\}$ be the polynomial system defined as in Eq. (6), where each coordinate of $\mathbf{s}_{\text{solve}}$ is restricted to an integer interval $\llbracket -T, T \rrbracket$. Let N_{sol} denote the number of solutions to $\mathcal{F}_{new}$. Then we have*

$$\mathbb{E}[N_{\text{sol}}] \approx ((2T + 1)^{n_{\text{solve}}} - 1) \cdot \left(1 - \left(1 - \frac{1}{q}\right)^K\right)^W + (1 - (1 - p_0)^K)^W.$$

Proof. Consider a single polynomial $F_j = \prod_{i=1}^K (b'_{j_i} - \langle \mathbf{a}'_{j_i}, \mathbf{s}_{\text{solve}} \rangle)$. Since every entry of $\mathbf{A}_{\text{solve}}$ and $\mathbf{b}$ is independent and uniformly random over $\mathbb{Z}_q$, for any fixed short vector $\mathbf{x}_{j_i}$ obtained from Algorithm 2, the two derived quantities

$$\mathbf{a}'_{j_i} = \mathbf{A}_{\text{solve}}^\top \mathbf{x}_{j_i} \quad \text{and} \quad b'_{j_i} = \langle \mathbf{b}, \mathbf{x}_{j_i} \rangle$$

are also independent and uniformly distributed over $\mathbb{Z}_q$. For a candidate solution $\tilde{\mathbf{s}} \neq \mathbf{s}_{\text{solve}}$, where each coordinate is chosen from the interval $\llbracket -T, T \rrbracket$, the probability that the linear term $b'_{j_i} - \langle \mathbf{a}'_{j_i}, \tilde{\mathbf{s}} \rangle$ equals zero is $1/q$. Hence,

$$\mathbb{P}[F_j(\tilde{\mathbf{s}}) = 0] = 1 - (1 - 1/q)^K.$$

Since the polynomials $F_1, F_2, \dots, F_W$ are built from disjoint sample sets, imply that a probability $\left(1 - (1 - 1/q)^K\right)^W$. Multiplying this probability by the number of candidates $\tilde{\mathbf{s}} \neq \mathbf{s}_{\text{solve}}$ and adding the probability that the true secret satisfies the system gives the desired expectation. $\square$

To successfully recover the secret, the polynomial system must simultaneously satisfy two competing requirements: it must contain the true solution with high probability while keeping the expected total number of solutions very small. Balancing these leads to the following constraints on the parameters K and W :

$$((2T + 1)^{n_{\text{solve}}} - 1) \cdot \left(1 - \left(1 - \frac{1}{q}\right)^K\right)^W + (1 - (1 - p_0)^K)^W \approx 1, \quad (8)$$

$$(1 - (1 - p_0)^K)^W \geq \theta, \quad (9)$$

where θ is a threshold that guarantees a high probability of correctness.

In practice, one can improve the reliability of recovering the true secret $\mathbf{s}_{\text{solve}}$ by independently constructing multiple polynomial systems of the form (6) and solving each one separately. All candidate solutions are then ranked by their frequency of occurrence, and those whose scores exceed a predefined threshold⁵ are retained. For each retained candidate, we substitute it back into the original LWE samples to obtain a sub-LWE problem of reduced dimension, solve

⁵ The threshold is determined by the probability parameter θ . For instance, with $\theta = 0.8$ and 8 independently constructed systems, the probability that the score of the true secret is at least 3 exceeds 99.8%; hence the threshold can be set to 3.

it to recover the remaining part of the secret, and finally verify the full candidate by checking whether the resulting error vector $\tilde{\mathbf{e}} = \mathbf{b} - \mathbf{A}\tilde{\mathbf{s}} \bmod q$ follows the expected distribution. Only candidates that pass this distributional test are accepted as valid solutions.

Remark 1. After the first stage, an exponential number of new LWE samples is available. One may filter them, retaining only those whose vectors $\mathbf{a}'$ share a common position of zero entries. This approach implicitly eliminates the corresponding variables—we let γ denote the number of variables removed in this way. However, such filtering shrinks the pool of usable samples, thereby limiting the feasible choices for the parameters K and W . Consequently, a careful balance must be struck between the number of variables eliminated γ and the number of samples retained.

We experimentally verify the correctness of the proposed polynomial system $\mathcal{F}_{\text{new}}$. The full verification codes are available on GitHub ⁶.

Experimental Verification of the Polynomial System Construction. To confirm the correctness, we verify two key statements: the probability that the true secret satisfies the system, and the expected number of solutions to the system. This amounts to checking Eq. (7) and Lemma 6. The verification employs the following two parameter sets:

- (1) $(q, \eta, n, m, n_{\text{lat}}, \beta_{\text{bkz}}, \beta_{\text{sieve}}) = (3329, 1, 50, 50, 40, 40, 80)$;
- (2) $(q, \eta, n, m, n_{\text{lat}}, \beta_{\text{bkz}}, \beta_{\text{sieve}}) = (3329, 1, 60, 60, 50, 40, 80)$.

Here $\eta = 1$ indicates that both the secret and the error are drawn from the centered binomial distribution $\mathcal{B}_1$. After dual lattice reduction each instance provides $N = 99438$ new samples.

The zero-error probability p_0 is computed via Eq. (3), yielding 0.005237 for Set (1) and 0.003761 for Set (2). For the two sets we use different construction parameters (K, W) :

$$\text{Set (1): } (K, W) = (700, 8), \quad \text{Set (2): } (K, W) = (1000, 9).$$

The corresponding theoretical probabilities that the true secret satisfies system $\mathcal{F}_{\text{new}}$ computed from Eq. (7) are both 0.81. The theoretical expected numbers of solutions computed from Lemma 6 are 0.91 for Set (1) and 1.13 for Set (2).

Verification of Eq. (7). For each set we generate 20 LWE instances. For each instance, knowing the true amplified error $\mathbf{e}'$, we perform 4000 independent simulation runs. In each run we draw K distinct samples without replacement over W trials ($K \cdot W \leq N$). A run is successful iff every trial contains at least one zero—this means the corresponding system would admit the true secret. The average success rate over the 4000 runs, denoted $\hat{p}_i^{(1)}$ and $\hat{p}_i^{(2)}$ for the i -th instance ($i = 1, \dots, 20$), gives the simulated probability.

⁶ <https://github.com/latticewalker/Algebraic-Analysis-for-LWE>.

For Set (1): mean $\bar{p}^{(1)} = 0.82$ (std 0.043). For Set (2): mean $\bar{p}^{(2)} = 0.80$ (std 0.037). Both are in close agreement with the theoretical value 0.81, supporting Eq. (7).

Verification of Lemma 6. Using the same 20 LWE instances per set and the same (K, W) as above, we independently construct and solve 100 systems $\mathcal{F}_{\text{new}}$ per instance. For each system we record the number of solutions. The per-instance average is denoted $\hat{\mu}_i^{(s)}$ ($s = 1, 2; i = 1, \dots, 20$); the overall mean $\bar{\mu}^{(s)}$ and standard deviation are:

- Set (1): $\bar{\mu}^{(1)} = 0.91$ (std 0.06), solution counts across all systems range from 0 to 3, true secret appears in 82.5% of systems.
- Set (2): $\bar{\mu}^{(2)} = 1.14$ (std 0.07), counts range 0–4, true secret appears in 82% of systems.

These empirical means are nearly identical to the theoretical expectations (0.91 and 1.13), and the variances and ranges are small. The high frequency of the true secret further matches the theoretical probability 0.81.

The results collectively validate Lemma 6, confirm the reasonableness of Eq. (7), and thereby verify the correctness of the proposed polynomial system $\mathcal{F}_{\text{new}}$.

Complexity of Solving Eq. (6) by the Resultant-based Method. To solve the system $\mathcal{F}_{\text{new}}$, we first consider the Gröbner basis methods. However, two main difficulties arise. First, it is unclear whether the conjecture 1, which underpins the “lowest achievable complexity estimates” from [32], still holds for our newly constructed system; consequently, the complexity analysis will likely need to rely on the semi-regularity assumption. Second, the polynomial degrees in our system remain relatively high—a property generally detrimental to efficient Gröbner basis computation. Therefore, we opt for a resultant-based method. This approach offers two key advantages in our context: (1) it yields a concrete complexity estimate for the second stage without relying on any heuristic assumptions, and (2) its complexity is primarily governed by the number of variables and the partial degree of each variable—quantities that we can control tightly.

Specifically, we introduce a polynomial constraint of the form for each variable

$$s_j \prod_{i=1}^T (s_j + i)(s_j - i) = s_j^d - h_j(s_j),$$

where $d = 2T + 1$ and $h_j(s_j)$ are polynomials of degree less than d . The **reduction system** is then defined as

$$\mathcal{P} = \{s_1^d - h_1(s_1), s_2^d - h_2(s_2), \dots, s_{n_{\text{solve}}}^d - h_{n_{\text{solve}}}(s_{n_{\text{solve}}})\}. \quad (10)$$

Whenever the degree of any variable in a polynomial reaches or exceeds d , we reduce it modulo the system $\mathcal{P}$, thereby ensuring that the degree of each variable remains strictly below d . The detailed reduction procedure is described in Algorithm 3, and its cost is analyzed in Lemma 7.

Algorithm 3 $\text{Reduce}_u(g(x_1, x_2, \dots, x_u), \mathcal{P})$

Input: A polynomial $g \in \mathbb{Z}_q[x_1, x_2, \dots, x_u]$, where $u \leq n$ and $\deg_{x_i}(g) \leq 2d - 2$ for $1 \leq i \leq u$, and a reduction system $\mathcal{P}$.

Output: A polynomial g such that the degree of each variable is less than d .

```

1: if  $u = 0$  then
2:   return  $g$ 
3: end if
4: Express  $g$  as  $g = \sum_{i=0}^{2d-2} g_i(x_1, x_2, \dots, x_{u-1})x_u^i$ .
5:  $\rho \leftarrow 0$ . ▷ Initialization
6: for  $i = 0$  to  $d - 2$  do
7:    $\rho \leftarrow \rho + \text{Reduce}_{u-1}(g_i, \mathcal{P}) \cdot x_u^i + \text{Reduce}_{u-1}(g_{i+d}, \mathcal{P}) \cdot (x_u^{i+d} \bmod (x_u^d - h_u))$ .
8: end for
9:  $\rho \leftarrow \rho + \text{Reduce}_{u-1}(g_{d-1}, \mathcal{P}) \cdot x_u^{d-1}$ . ▷ Total of  $2d - 1$  calls to  $\text{Reduce}_{u-1}$ 
10: return  $\rho$ 

```

Lemma 7. Let $C(u, d)$ denote the worst-case time complexity of Algorithm 3 in $\mathbb{Z}_q$ -operations. Then

$$C(u, d) = \mathcal{O}((2d - 1)^{u+1}).$$

Proof. When $u = 0$, the complexity is approximately 0. For $u \geq 1$, the algorithm makes $2d - 1$ recursive calls to Reduce_{u-1} and performs $d - 1$ multiplications involving the pre-computed $x_u^{i+d} \bmod (x_u^d - h_u)$. Each $\text{Reduce}_{u-1}(g_{i+d}, \mathcal{P})$ contains at most d^{u-1} terms, so each multiplication costs $\mathcal{O}(d^u)$. This yields the recurrence

$$C(0, d) = 0, \quad C(u, d) = (2d - 1)C(u - 1, d) + (d - 1)d^u.$$

Expanding the recurrence gives

$$C(u, d) = (d - 1) \sum_{i=0}^{u-1} (2d - 1)^i d^{u-i} < \sum_{i=0}^{u-1} (2d - 1)^i d^{u-i+1} = \mathcal{O}((2d - 1)^{u+1}).$$

□

Theorem 2 provides the time complexity of solving $\mathcal{F}_{\text{new}}$ via the resultant-based method.

Theorem 2. Let $q \geq 2$, $m = n \geq 2$ be LWE parameters, with the secret $\mathbf{s}$ and error $\mathbf{e}$ having coordinates chosen from $\llbracket -T, T \rrbracket$. After performing dual lattice reduction with parameters n_{lat} , $n_{\text{solve}} = n - n_{\text{lat}}$, β_{bkz} , and β_{sieve} , let $\mathcal{F}_{\text{new}} = \{F_1, F_2, \dots, F_W\}$ be the polynomial system constructed according to Eq. (6) and satisfying the two constraints given in Eqs. (8) and (9). Then, with a success probability of at least θ , $\mathcal{F}_{\text{new}}$ can be solved by the resultant-based method, and its time complexity (in $\mathbb{Z}_q$ -operations) is upper bounded by

$$\mathcal{O}\left((2d - 1)!(d - 1) + W2^{2d-1}(d - 1) \cdot (v \log v \log \log v + (2d - 1)^{n_{\text{solve}}})\right), \quad (11)$$

where $d = 2T + 1$ and $v = (2d - 1)^{n_{\text{solve}}-1}$.

Proof. Set the parameter θ , then let K and W satisfy Eqs. (8) and (9), and let $\mathcal{P}$ be the reduction system defined in Eq. (10). The overall complexity of solving $\mathcal{F}_{\text{new}}$ can be separated into three parts: (1) the cost of expanding and reducing the polynomials during the construction of the system, (2) the cost of the successive resultant-based elimination, and (3) finding roots of the univariate polynomials. For the worst-case complexity estimate, we assume that every polynomial contains the maximal possible number of monomials.

First, during the construction of $\mathcal{F}_{\text{new}}$, Algorithm 3 is applied whenever any variable degree reaches d , ensuring that all degrees remain below d . In the subsequent resultant computation, the same algorithm is invoked after each polynomial multiplication, which can guaranty that no intermediate polynomial contains a variable of degree exceeding $2d - 2$. Under the worst-case, the costs of reductions during the construction $\mathcal{F}_{\text{new}}$ are negligible compared with the resultant computation.

Second, let $H_{n_{\text{solve}},j} = F_j$ for $j = 1, 2, \dots, W$. The elimination process then proceeds iteratively as follows:

$$\left\{ \begin{array}{l} H_{i-1,1} = \text{Res}_{s_i}(H_{i,1}, s_i^d - h_i(s_i)), \\ H_{i-1,2} = \text{Res}_{s_i}(H_{i,2}, s_i^d - h_i(s_i)), \\ \vdots \\ H_{i-1,W} = \text{Res}_{s_i}(H_{i,W}, s_i^d - h_i(s_i)), \end{array} \right. \quad (12)$$

where $i = n_{\text{solve}}, n_{\text{solve}} - 1, \dots, 2$. Because Algorithm 3 is applied after every multiplication, the degree of each variable never exceeds $d - 1$; consequently, each resultant is computed as a $(2d - 1)$ -order determinant. A key advantage of this construction is that the bottom $d - 1$ rows of the determinant contain only numbers, which requires only a small number polynomial multiplications and simplifies the computation. Since eliminating $s_{n_{\text{solve}}}$ asymptotically dominates⁷, we only focus on its cost, which consists of two parts: (i) the cost of a single polynomial multiplication followed by degree reduction, and (ii) the total number of such multiplication-reduction steps.

From Lemma 1, the multiplication of two polynomials in $n_{\text{solve}} - 1$ variables, each of partial degree at most $d - 1$, requires $\mathcal{O}(v \log v \log \log v)$ operations in $\mathbb{Z}_q$, where $v = (2d - 1)^{n_{\text{solve}} - 1}$. According to Lemma 7, subsequent reduction via Algorithm 3 adds an additional cost of $\mathcal{O}((2d - 1)^{n_{\text{solve}}})$. Hence, one multiplication-reduction step costs $\mathcal{O}(v \log v \log \log v + (2d - 1)^{n_{\text{solve}}})$.

In each resultant computation, the $(2d - 1)$ -order determinant has the following structure: The first d rows are populated by multivariate polynomials

⁷ In practice, eliminating the first few variables is relatively fast due to the sparsity of the polynomial terms; the process slows down as terms become denser with successive eliminations. However, precisely tracking the term counts during each elimination step is highly complex. Hence, this paper adopts an upper-bound estimate by assuming that all potential monomials appear during the first elimination. A more precise complexity analysis is left for future work.

and zeros, while the last $d - 1$ rows contain only numbers. Treating each distinct polynomial as a distinct variable, the expansion of this determinant yields monomials of degree d . The cost of this expansion is $\mathcal{O}((2d - 1)!(d - 1))$, and the number of such monomials is bounded by $\binom{2d-1}{d} < 2^{2d-1}$. Hence, the total number of multiplications required per resultant is at most $2^{2d-1} \cdot (d - 1)$.

Finally, the cost of eliminating $s_{n_{\text{solve}}}$ is at most

$$\mathcal{O}\left((2d - 1)!(d - 1) + W2^{2d-1}(d - 1) \cdot (v \log v \log \log v + (2d - 1)^{n_{\text{solve}}})\right).$$

The costs of eliminating the remaining variables and the univariate polynomial root-finding (see Lemma 2 in Section 2.5 for details) are asymptotically negligible. $\square$

The time complexity above is expressed in terms of the number of field operations. To obtain a bit complexity, one can equate one field operation to $\lceil \log q \rceil^2$ bit operations.

We compare Algorithm 1 with the “lowest achievable complexity estimate” from [32] using CRYSTALS-KYBER as a concrete example. We first derive the time complexity of our algorithm and then describe the parameter selection that minimizes it.

The overall complexity combines the costs of the two stages. The first stage, from Lemma 5, costs $\mathcal{O}(2^{0.292\beta})$, where β is the blocksize. For the second stage, we start from Eq. (11) of Theorem 2. Since the secret follows a centered binomial distribution $\mathcal{B}_\eta$ ($\eta = 3$ for KYBER-512, $\eta = 2$ for KYBER-768/1024), the parameter T and thus $d = 2T + 1$ are small, making $(2d - 1)! \cdot (d - 1)$ negligible. We refine the estimate by treating each distinct polynomial appearing in the resultant determinant as a separate variable, expanding the determinant in practice, counting the resulting monomials, and thereby obtaining the exact number of multiplications: 2832 for KYBER-512 and 152 for KYBER-768/1024. The combined time complexity can be estimated as

$$\mathcal{O}\left(2^{0.292\beta} + \lceil \log q \rceil^2 \cdot W \cdot w_i \cdot (v \log v \log \log v + (2d - 1)^{n_{\text{solve}} - \gamma})\right),$$

where $d = 2T + 1$, $v = (2d - 1)^{n_{\text{solve}} - 1 - \gamma}$, $w_1 = 2832$, $w_2 = 152$, and γ is the number of variables eliminated via Remark 1.

We fix the overall success probability at 0.8. To minimize this complexity, we perform a parameter search under the following procedure. The blocksize parameter β (where $\beta = \beta_{\text{bkz}} = \beta_{\text{sieve}}$) is chosen according to the core-SVP methodology in [31, Table 4], ensuring it remains strictly below the corresponding primal-attack BKZ blocksize of 403, 625, and 877 for KYBER-512/768/1024, respectively. We then search for the parameter tuple $(n_{\text{lat}}, K, W, \gamma)$ that yields the lowest complexity while satisfying Eqs. (8), (9) and the sample-pool bound $K \cdot W < N/q^\gamma$.

The resulting time complexity estimates for CRYSTALS-KYBER are listed in Table 1.

Table 1. Complexity comparison between Algorithm 1 and the “lowest achievable estimates” from [32] for CRYSTALS-KYBER. The parameter d_{reg} is the lowest achievable degree of regularity, computed via Eq. (4). All complexities are expressed in bits ($\log_2$).

Scheme	Complexity		d_{reg}	Key Parameters						
	[32]	Ours		n_{lat}	β_{bkz}	β_{sieve}	γ	K	W	
KYBER-512	1584	709	306	345	395	395	0	$2^{16.94}$	$2^{62.85}$	
KYBER-768	1588	962	232	492	600	600	3	$2^{17.14}$	$2^{71.39}$	
KYBER-1024	2014	1311	287	644	850	850	5	$2^{17.62}$	$2^{96.56}$	

4 Further Discussion on the Resultant-Based Method

Gröbner bases constitute a general algebraic solving tool, first applied to the Arora-Ge system by Albrecht et al. [2]. That early work, however, relied on the semi-regularity assumption. More recently, Steiner [32] provided a proven method for solving the Arora-Ge system using Gröbner bases. As their analysis of KYBER-768 showed, the proven bound obtained in this way is relatively loose and leaves a significant gap compared to the “lowest achievable estimate” reported in [32]. To partially bridge this gap, we introduce a resultant-based method. In this section, we demonstrate that this resultant method not only synergizes with the dual lattice reduction stage but also stands as an independent algebraic solver of broader interest.

We focus on LWE with a small secret, meaning each coordinate of the secret vector is chosen from a small, fixed interval. Without loss of generality, we assume each secret coordinate is sampled from the integer interval $\llbracket -T, T \rrbracket$ and the error is chosen from a larger interval.

Theorem 3 establishes a provable complexity estimate for solving the Arora-Ge system via the resultant-based method; its proof follows directly from Theorem 2. In the small-secret setting, this bound yields an **exponential speed-up** over the proven estimate in [32]. Hence, the resultant-based method constitutes a practical and competitive alternative to Gröbner bases for the Arora-Ge system, thereby broadening the algebraic cryptanalytic toolkit for LWE.

Theorem 3. *Let $q \geq 2$, $m = n \geq 2$ be LWE parameters, with the secret $\mathbf{s}$ having coordinates chosen from a small interval $\llbracket -T, T \rrbracket$. Let $\mathcal{F}_{\text{AG}}$ be the Arora-Ge system. Then the time complexity (in $\mathbb{Z}_q$ -operations) of solving $\mathcal{F}_{\text{AG}}$ by the resultant-based method is upper bounded by*

$$\mathcal{O}\left((2d-1)!(d-1) + m2^{2d-1}(d-1) \cdot (v \log v \log \log v + (2d-1)^n)\right). \quad (13)$$

where $d = 2T + 1$.

Proof. For the worst-case estimate under the elimination scheme of Eq. (12) (Theorem 2), we also concentrate on the cost of eliminating the first variable. The

analysis requires determining two quantities: the cost of a single multiplication-reduction step and the total number of such steps. Eliminating the first variable involves at most $m \cdot 2^{2d-1}(d-1)$ such steps, each costing

$$\mathcal{O}(v \log v \log \log v + (2d-1)^n),$$

where $v = (2d-1)^{n-1}$. So the complexity can be estimated by

$$\mathcal{O}\left((2d-1)!(d-1) + m2^{2d-1}(d-1) \cdot (v \log v \log \log v + (2d-1)^n)\right).$$

□

Under the small-secret setting, we simplify Eq. (13) to a reasonable extent to facilitate a clear complexity comparison. Because T is small, the term $(2d-1)!(d-1)$ is negligible, which yields a cleaner expression:

$$\mathcal{O}\left(m \cdot n \log n \cdot 2^{2d-1+\log(d-1)+\log(2d-1) \cdot n}\right). \quad (14)$$

Theorem 22 of [32] gives the proven complexity for the same small secret setting,

$$\mathcal{O}\left(m \cdot (d-1) \cdot n^2 \cdot 2^{\omega \cdot 2^{\frac{1}{\ln(2)}} \cdot (d-1)^{1-\frac{1}{\ln(4)}} \cdot n^{2-\frac{1}{\ln(4)}}}\right). \quad (15)$$

The dominant term in Eq. (14) scales as 2^n , while the corresponding term in Eq. (15) scales as $2^{n^{2-1/\ln 4}}$. Since $2-1/\ln 4 > 1$, this alone implies an exponential speed-up of our bound over the proven estimate of [32]. Moreover, comparing the coefficients, one can easily verify that $\omega \cdot 2^{1/\ln 2} \cdot (d-1)^{1-1/\ln 4} > \log(2d-1)$ holds for all integers $d \geq 2$, which further substantiates the significant advantage of our method.

As a concrete example, for CRYSTALS-KYBER, we use a more precise complexity formula instead of the asymptotic form Eq. (14):

$$\mathcal{O}([\log q]^2 \cdot m \cdot w_i \cdot (v \log v \log \log v + (2d-1)^n)), \quad (16)$$

where $d = 2T + 1$, $v = (2d-1)^{n-1}$, and the scheme-dependent constants are $w_1 = 2832$ for KYBER-512 and $w_2 = 152$ for KYBER-768/1024.

Table 2 presents the complexity estimates obtained via Eq. (16), alongside the corresponding proven estimates from [32].

5 Augmenting Algorithm 1 with Hints

At CRYPTO 2025, Cao et al. [9] introduced a general framework that converts arbitrary numbers of perfect, modular, and approximate hints into a lattice problem, together with a complexity estimator for solving LWE with such auxiliary information. The three types of hints are formally defined as

Perfect hint: $\langle \mathbf{s}, \mathbf{v} \rangle = l \in \mathbb{Z}_q$,

Modular hint: $\langle \mathbf{s}, \mathbf{v} \rangle \equiv l \pmod k$,

Approximate hint: $\langle \mathbf{s}, \mathbf{v} \rangle + e_\sigma = l \in \mathbb{Z}_q$.

Table 2. Complexity of solving the Arora-Ge system using resultant-based method for CRYSTALS-KYBER. Both the proven and optimistic estimates from [32] are provable and are included for comparison. All costs are measured in bits ($\log_2$).

Scheme	Our Method	Proven estimates [32]	Optimistic estimates [32]
KYBER-512	1933	4249	4479
KYBER-768	2471	5554	4717
KYBER-1024	3283	7402	6030

In this section, we systematically examine how side information—both individual hints and their combinations—affects the performance of Algorithm 1. We first analyze the impact of each hint type on the algorithm’s complexity, and then explore the effect of combining multiple hints, which is of particular importance in practice.

Perfect hints. Adding a perfect hint is straightforward: it introduces a linear equation that directly eliminates one variable from the polynomial system.

Modular hints. A modular hint of the form $\langle \mathbf{s}, \mathbf{v} \rangle - l \in \Omega \subseteq \mathbb{Z}_q$ yields a polynomial whose roots lie in Ω . For $\Omega = \{0, 1\}$ this gives $(\langle \mathbf{s}, \mathbf{v} \rangle - l)(\langle \mathbf{s}, \mathbf{v} \rangle - l - 1) = 0$. Adding such constraints to $\mathcal{F}_{\text{new}}$ further shrinks the candidate set, enabling larger n_{lat} and better complexity. A full treatment of modular hints is quite complicated; we instead focus on the special case $s_i \equiv 0 \pmod t$ for integer $t < q$.

Approximate hints. One important class of approximate hints consists of Hamming-weight hints, which reveal the Hamming weight of secret coordinates. For example, given $s_1 \in \{-2, -1, 0, 1, 2\}$ and the knowledge that $H(s_1) = 1$, the candidate set is reduced to $\{1, 2\}$. Such a reduction decreases the degree of the corresponding variable in $\mathcal{P}$, directly benefiting the resultant-based method.

The following examples illustrate how hints, individually or in combination, directly benefit Algorithm 1.

Example 1 (Hamming hints). Suppose the secret coordinates lie in $\llbracket -2, 2 \rrbracket$. If $H(s_i) = 1$ (resp. 2), then $s_i \in \{1, 2\}$ (resp. $\{-1, -2\}$). The degree- d polynomial of $\mathcal{P}$ defined in Eq. (10) can thus be replaced by the quadratic $s_j^2 - 3s_j + 2$ or $s_j^2 + 3s_j + 2$; the degree of each variable in $\mathcal{F}_{\text{new}}$ is then bounded by 2.

Example 2 (Perfect + Hamming hints). Assume we have n_1 perfect hints and that the Hamming weight of the secret is known to be 1. The perfect hints eliminate n_1 variables, and the remaining analysis follows Example 1.

Example 3 (Modular + Hamming hints). Consider m LWE samples $(\mathbf{a}_i, \mathbf{b}_i)$ with a secret $\mathbf{s}$ whose coordinates lie in $\llbracket -5, 5 \rrbracket$. Suppose we are given both Hamming-weight hints (indicating $H(s_i) = 1$) and modular hints requiring $s_i \equiv 0 \pmod 2$.

Together, these constraints reduce the number of possible values for each coordinate from 11 to 2, and the subsequent analysis proceeds analogously to Example 1.

6 Conclusions

This work introduces a new algebraic algorithm for LWE that strategically integrates dual lattice reduction into algebraic analysis for LWE. We also demonstrate that the classical Arora-Ge polynomial arises as a special case of our more general construction, thereby revealing a unifying algebraic perspective.

The proposed algorithm consists of two distinct stages. First, a dual lattice reduction stage transforms the original LWE samples into a set of dimension-reduced samples with amplified error. Second, instead of applying Gröbner basis methods to the Arora-Ge system, we construct a new polynomial system tailored to the distribution of the amplified error and solve it via a resultant-based method. This two-stage design is pivotal, as our polynomial construction exhibits improved compatibility with the preceding reduction stage, allowing us to effectively balance the trade-off between dimension reduction and error amplification. Concretely, for the NIST selected algorithm CRYSTALS-KYBER, our algorithm delivers complexity estimates that are better than the results of solving the Arora-Ge system with Gröbner bases reported in [32].

Independently, this work establishes the resultant-based method as a new, competitive algebraic solver in its own right. When applied directly to the original Arora-Ge system (without dual lattice reduction) in the small secret LWE setting, it delivers a provable bound that exhibits an exponential speed-up over the proven estimates in [32]. We further illustrate how various forms of side information—including perfect, modular, and approximate hints—can be naturally incorporated into the polynomial system to accelerate the our two-stage algorithm.

Our work explores whether techniques from dual attacks can meaningfully enhance the algebraic analysis of LWE, thereby narrowing the gap between these two cryptanalytic approaches. The primary goal is not to challenge the established security estimates of standardized LWE-based schemes, but to investigate potential synergies between different analytical paradigms. Although the complexity estimates we obtain for KYBER-512/768/1024 remain substantially higher than those of the best primal or dual attacks, they represent a significant improvement over previous algebraic results. This progress confirms the viability of integrating lattice reduction into the algebraic cryptanalysis framework.

During the course of this research, we identified several promising avenues for advancing the algebraic analysis of LWE:

Open Question 1: Fast Solvability Testing via Resultant. Empirical results in Section 3.2.2 indicate that a the system $\mathcal{F}_{\text{new}}$ is very likely to have no solution if it does not contain the true secret. This raises a natural question: can the resultant be leveraged to develop a fast, dedicated test for the solvability of

such systems? If feasible, one could set the probability that a system contains the true secret to a very low value, repeat the construction many times, and solve only those systems that pass the solvability test. This approach could lead to a substantial reduction in the overall computational cost.

Open Question 2: Leveraging Additional Samples. Our analysis assumes the standard setting $m = n$. Two natural extensions arise: whether and how extra samples can improve efficiency, and how many samples are needed for the algebraic approach to meet the NIST security levels for CRYSTALS-KYBER. Both questions are of direct relevance from a designer’s worst-case perspective.

Open Question 3: Quantum Acceleration. As post-quantum cryptographic algorithms must resist both classical and quantum adversaries, exploring quantum variants of our algorithm is essential. Whether quantum techniques can offer a substantial asymptotic speedup remains an important long-term research direction.

In summary, this paper helps bridge the gap in which algebraic analysis lagged far behind dual attacks for LWE, offering improved complexity estimates and presenting a refined algebraic solver. We hope it stimulates further work on the algebraic analysis of LWE.

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